ST MARY'S DSG, KLOOF

MATHEMATICS PAPER 2 -edited

GRADE 10 | 19 NOVEMBER 2019

TIME: 2 HOURS | TOTAL: ~~120~~ 112 MARKS

NAME: Memo | MATHS TEACHER: _____

READ THE FOLLOWING INSTRUCTIONS CAREFULLY:

1. This question paper consists of 17 pages and 17 questions.
2. Formulae have been provided below.
3. There are 4 sections. Write your name and your Mathematics teacher's name on each section and hand in separately.
4. Answer all the questions.
5. Diagrams are not necessarily drawn to scale.
6. **All the necessary working details must be clearly shown.**
7. **Reasons must be given unless specified otherwise.**
8. Approved non-programmable calculators may be used except where otherwise stated. Ensure that your calculator is in degree mode.
9. It is in your own interest to write legibly and to present your work neatly.
10. Answers may be left in surd format or rounded off to **two decimal places**.

FORMULAE

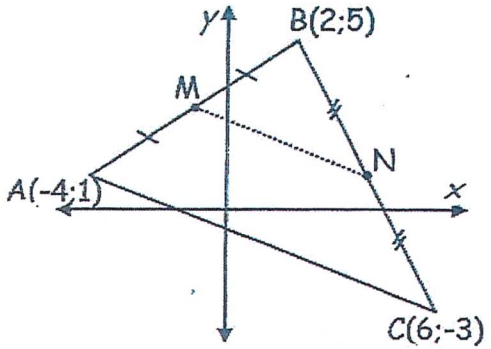
$y = mx + c$ $\quad m = \frac{y_2 - y_1}{x_2 - x_1}$ $\quad d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$ $\quad m = \tan\theta$

$M = \left(\frac{x_1 + x_2}{2}; \frac{y_1 + y_2}{2}\right)$ $\quad \tan\theta = \frac{\sin\theta}{\cos\theta}$ $\quad \sin^2\theta + \cos^2\theta = 1$

$A = \pi r^2$ $\quad V = Area\ of\ base \times Height$

Question 1

In the diagram below, A(-4;1), B(2;5) and C(6;-3) are the coordinates of ΔABC.
M and N are the midpoints of AB and BC respectively.



a) Give the co-ordinates of M (3)

$M\left(\frac{-4+2}{2}; \frac{1+5}{2}\right) = M(-1;3)$

b) Determine the equation of the straight line BC (4)

$m = \frac{5-(-3)}{2-6}$ $\quad y = -2x + c$

$m = -2$ $\quad 5 = -2(2) + c$

$9 = c$

$\therefore y = -2x + 9$

c) Calculate the length of AC (3)

$AC = \sqrt{(-4-6)^2 + (1-(-3))^2}$

$= \sqrt{(-10)^2 + (4)^2}$

$= \sqrt{116}$

OR 10,77 units

OR $2\sqrt{29}$

d) Explain clearly why $B\hat{M}N = B\hat{A}C$. (3)

$BM = MA$; $BN = NC$

$\therefore MN \parallel AC$ (midpt th)

$\therefore B\hat{M}N = B\hat{A}C$ (corresp ∠s: MN∥AC)

OR

$\frac{BM}{MA} = \frac{BN}{NC} = \frac{1}{1}$

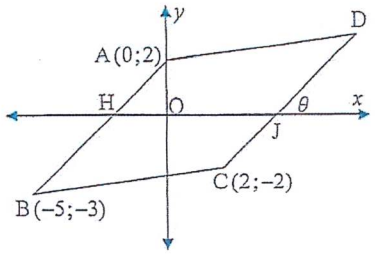
$\therefore MN \parallel AC$ (converse Prop Int. th)

[13]

Question 2

A(0;2), B(−5;−3), C(2;−2) and D are the vertices of parallelogram ABCD.

The x-intercept of AB is point H.
CD cuts the x-axis at J and makes an angle of θ, as indicated, with the x-axis.



a) Determine the gradient of AB. (2)

$m = \frac{2--3}{0--5}$

$m = 1$

b) Calculate the size of θ correct to one decimal place. (3)

$mCD = mAB = 1$ opp sides parm //

$\therefore \tan\theta = 1$

$\theta = 45°$

c) Determine the value of the x-intercept at H. (3)

$H(x;0) \quad \therefore \quad 1 = \frac{2-0}{0-x}$

$-x = 2$

$x = -2$

d) Determine the coordinates of D. (2)

$BC // AD \quad \therefore (x;y) \rightarrow (x+7 ; y+1)$

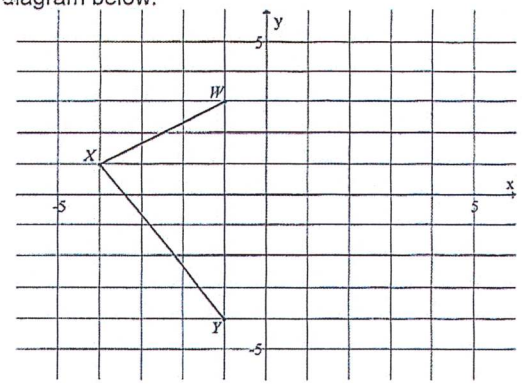
$A \rightarrow D \quad : (0;2) \rightarrow (0+7 ; 2+1)$

$\therefore D(7;3)$

[10]

Question 3

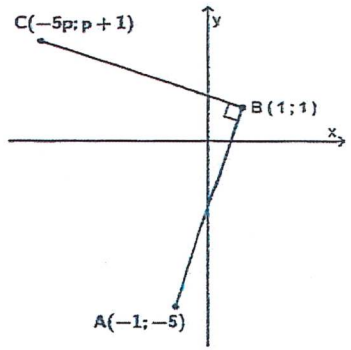
a) Refer to the diagram below.



Give the co-ordinates of the point Z, so that WXYZ is a kite. Z (2;1) (2)

b) In the diagram alongside, which is not drawn to scale, $AB \perp BC$.
Points $A(-1;-5)$, $B(1;1)$ and $C(-5p;p+1)$ are shown.

Determine the value of p.



$m AB = \frac{-5-1}{-1-1} = 3$

$\therefore mCB = -\frac{1}{3}$

$-\frac{1}{3} = \frac{1-(p+1)}{1+5p}$

$\frac{-1}{3} = \frac{1-p-1}{1+5p}$

$-1-5p = -3p$

$-1 = 2p$

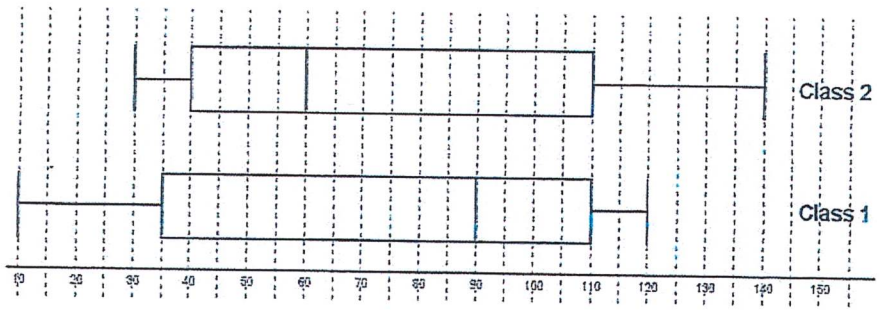
$-\frac{1}{2} = p$

(5)

[7]

Question 5

The exam results of 2 Maths classes were recorded and shown in the box and whisker diagrams below. The exam total is 150 marks.



State whether the following statements are True or False.

a) The range of the classes ~~are equal~~. T (1)

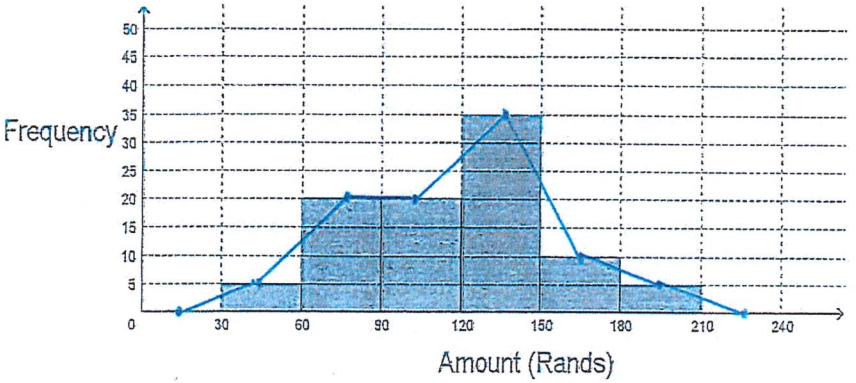
b) Most results in class 1 lie ~~between~~ 35 and 90. F (1)

c) The interquartile range for class 2 is 70. T (1)

[3]

Question 6

The Histogram below shows the amount of money, in Rands, that a Grade 10 group raised at their Entrepreneur Day.



a) Complete the table below from the data given in the Histogram. (1)

Amount (x) in Rands	Frequency
$30 < x \leq 60$	5
$60 < x \leq 90$	20
$90 < x \leq 120$	20
$120 < x \leq 150$	35
$150 < x \leq 180$	10
$180 < x \leq 210$	5
Total:	95

b) Use a ~~coloured~~ pen to draw a frequency polygon over the ~~Histogram~~. (2)

c) Determine the ~~modal~~ interval. $120 < x \leq 150$ (1)

d) Determine into ~~which~~ interval the lower quartile (Q_1) falls. $60 < x \leq 90$ (2)

e) Calculate ~~the approximate~~ mean amount of money raised by a ~~Grade~~ 10 pupil. Show all ~~working~~ details.

$\bar{x} = [45x5 + 75x20 + 105x20 + 135x35 + 165x10 + 195x5] \div 95$

$= 11175 \div 95$

$= 117,63$

(4)

[10]

Question 7

Six data values are listed in ascending order:

$$a\ ;\ b\ ;\ 7\ ;\ c\ ;\ 11\ ;\ 17$$

Determine the values of a, b and c, if the following further information is given regarding the data: (The mean value is represented by $\bar{x}$)

The range $=13$. $\bar{x}=\frac{28}{3}$ and $Q_2=8,5$

Range:

$17-a=13$

$4=a$

Q_2 : $\frac{7+c}{2}=8,5$

$7+c=17$

$c=10$

Mean:

$\frac{4+b+7+10+11+17}{6}=\frac{28}{3}$

$\frac{49+b}{6}=\frac{28}{3}$

$49+b=56$

$b=7$

[6]

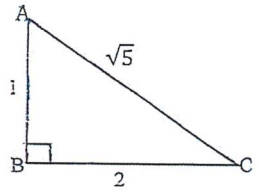
Question 8

a) Use your calculator to evaluate correct to 2 decimal places:

1) $\cos 66,6^\circ+1$ 1,40 (1)

2) $\sin^2 20^\circ$ 0,12 (1)

b) Use the diagram below to complete the following:

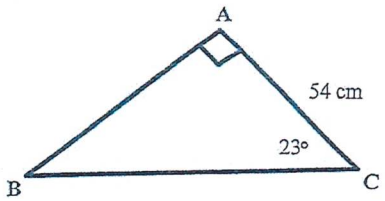


1) cos C = $\frac{2}{\sqrt{5}}$ (1)

2) sin A = $\frac{2}{\sqrt{5}}$ (1)

3) tan A = 2 (1)

c) Determine the length of BC, correct to two decimal places. (3)



$\cos 23^\circ=\frac{54}{BC}$

$BC=\frac{54}{\cos 23}$

$BC=58,66$ cm

[8]

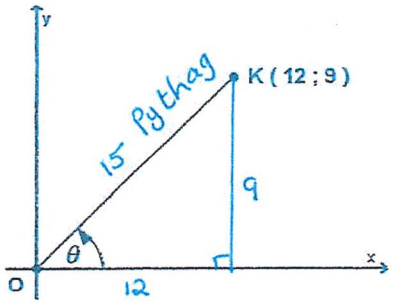
SECTION C

NAME:_____ MATHS TEACHER:_____

Question 9

K is the point $(12;9)$ and $K\hat{O}X = \theta$

Use the diagram to answer the questions that follow:



a) Determine, without the use of a calculator:

1) $\tan\theta$ (1)

$= \frac{9}{12}$

2) $3\cos\theta$ (3)

$OK^2 = 9^2 + 12^2$

$OK = 15$ (Pythagoras)

$3 \times \frac{12}{15}$

$= \frac{12}{5}$

3) the value of m if $\sin\theta = \frac{2}{m}$ (3)

$\frac{9}{15} = \frac{2}{m}$

$9m = 30$

$m = \frac{10}{3}$

b) Use a calculator to find the value of θ in the above diagram. (2)

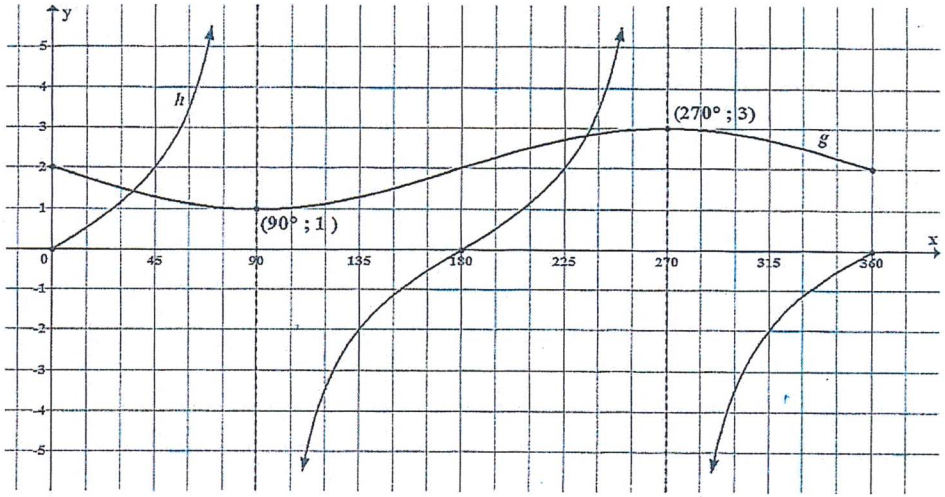
$\tan\theta = \frac{9}{12}$

$\theta = 36,87°$

[9]

Question 10

Sketched below are the graphs of $h(x) = 2\tan x$ and $g(x) = b\sin x + c$ for $x \in [0°; 360°]$.



(a) Determine the values of b and c.

$b = -1 \quad c = 2$ (2)

(b) Write down the period of h. $180°$ (1)

(c) Write down the equation of one of the asymptotes of h. $x = 90°$ or $270°$ (1)

(d) Write down one value of x for which $g(x) - h(x) = 2$. $x = 0$ or $x = 360°$ $x = 180°$ (1)

(e) Use the graph to determine the values of x for which $h(x) \geq 2$

$45° \leq x < 90° \; ; \; 225° \leq x < 270°$ (4)

OR $x \in [45°; 90°) \; ; \; x \in [225°; 270°)$

[9]

Question 11

a) Without using a calculator, evaluate: $\dfrac{\sin 30^o - \tan 45^o}{\cos 30^o \times \sin 90^o}$ (3)

$$\frac{\frac{1}{2} - 1}{\frac{\sqrt{3}}{2} \times 1}$$

$$= -\tfrac{1}{2} \div \tfrac{\sqrt{3}}{2} = -\tfrac{1}{2} \times \tfrac{2}{\sqrt{3}}$$

$$= \frac{-1}{\sqrt{3}}$$

b) Use the fundamental identities to simplify the following to one trigonometry ratio:

$\tan\theta \cdot \sin\theta + \cos\theta$ (3)

$$\frac{\sin\theta}{\cos\theta} \cdot \sin\theta + \cos\theta$$

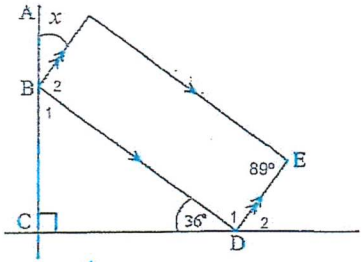
$$= \frac{\sin^2\theta + \cos^2\theta}{\cos\theta}$$

$$= \frac{1}{\cos\theta}$$

[6]

Question 12

a) Calculate, with reasons, the size of the angle marked x.



$\hat{B}_1 = 54°$ (∠s of Δ)

$\hat{B}_2 = 89°$ (opp ∠s parm)

$x = 37°$ (str line)

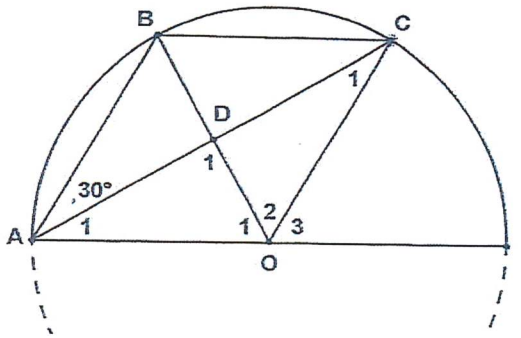
(3)

b) In the diagram below, O is the centre of the semi-circle.

A, B and C are on the circumference of the semi-circle.

ABCO is a **RHOMBUS**.

OB and AC intersect at D.



Determine, with reasons, the size of:

1) $\hat{C}_1$ = 30° (diags of rhombus bisect int ∠s) OR (alt ∠s : AB // CO) (2)

2) $\hat{D}_1$ = 90° (diags of rhombus ⊥) (2)

3) $\hat{O}_2$ = 60° (∠s of Δ DCO) (2)

[9]

SECTION D

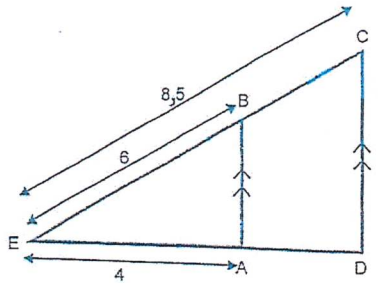
SECTION D

NAME:_____ MATHS TEACHER:_____

Question 13

a) In $\Delta CED, AB // DC$; $EC = 8,5\ cm$; $EB = 6\ cm$; $EA = 4\ cm$

Determine the length of AD.



$\frac{EB}{BC} = \frac{EA}{AD}$ (Prop int th BA // DC)

$\frac{6}{2,5} = \frac{4}{AD}$

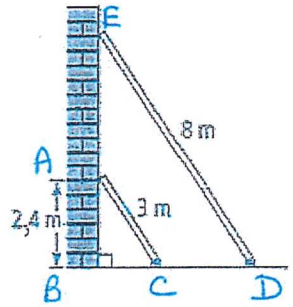
$AD = \frac{4 \times 2,5}{6}$

$AD = \frac{5}{3}$

(4)

b) Two extension ladders are leaning at the same angle, against a vertical wall.

The 3 metre ladder reaches 2,4 metres up the wall.



1) Prove that the two right-angled triangles are similar.

You will need to label the diagram. (2)

1. $\hat{A} = \hat{E}$ } corresp $\angle$s: ED // AC
2. $\hat{C} = \hat{D}$ }
3. $\hat{B} = 90°$ common

$\Delta ABC /// \Delta EBD$ (AAA)

2) How much further up the wall does the 8 metre ladder reach? Show clear reasoning.

$\frac{AB}{EB} = \frac{AC}{ED} = \frac{BC}{BD}$ Sides in Prop

$\frac{2,4}{EB} = \frac{3}{8}$

$EB = \frac{32}{5}$

$\therefore AE = \frac{32}{5} - 2,4$

$= 4m$

(4)

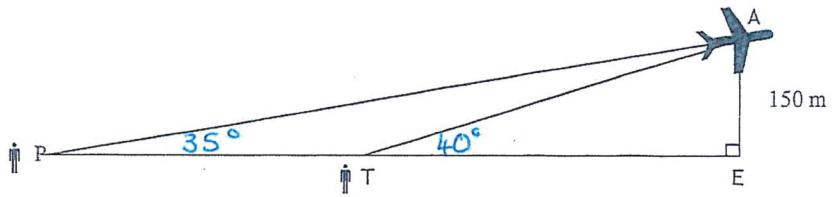
[10]

Question 14

Peter (P) and Tom (T) are observing an aeroplane in flight. From where Peter is standing he observes that the angle of elevation is 35°. Tom (T) notes that the aeroplane's angle of elevation was 40° from where he is standing.

At position E, the altitude of the plane was 150 m above the ground.

How far apart (length PT) are the two boys standing?



$\tan 35° = \frac{150}{PE}$

$PE = \frac{150}{\tan 35°}$

$PE = 214,22m$

$\tan 40° = \frac{150}{TE}$

$TE = \frac{150}{\tan 40}$

$TE = 178,76m$

$PT = 214,22 - 178,76$

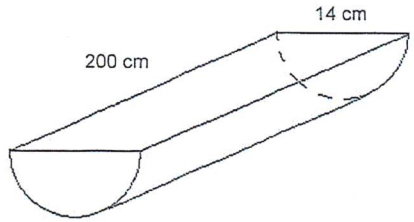
$= 35,46\ m$

[5]

Question 15

The diagram represents a water trough.

The length of the trough is 200 cm and the width at the top is 14 cm.



a) Calculate the volume of water that the trough can hold. (3)

$Vol = \frac{1}{2} \pi r^2 h$

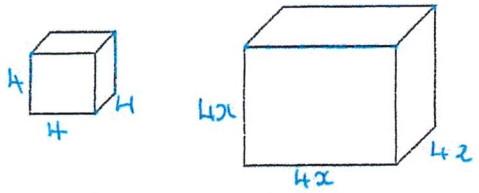
$= \frac{1}{2} \pi (7)^2 \times 200$

$= 4900\pi$

$= 15393,80\ cm^3$

b) A cube has a volume of 64 cm³. If each side is scaled up by a factor of x, determine the volume of the bigger cube in terms of x. (2)

(Diagrams are not drawn to scale!)



$Vol = (4x)^3$

$= 64x^3$

[5]

Question 16

A set of data has a mean value of 52,1 units. A box and whisker diagram has been drawn based on this data set:



a) Is the data symmetrical, positively skewed or negatively skewed? (1)

Negatively skewed

b) If each data value in the set is multiplied by 3 and then has 5 added to it, determine which one of the following statements would be true for the new data set:

Circle your choice.

A $\bar{x} = 161,3$ and IQR = 47

B $\bar{x} = 156,3$ and IQR = 42

(C) $\bar{x} = 161,3$ and IQR = 42 → $(\bar{x} \times 3) + 5$, IQR × 3

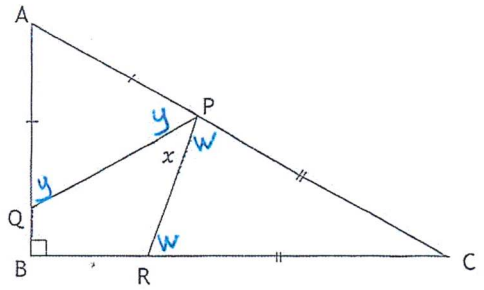
D $\bar{x} = 156,3$ and IQR = 14

(2)
[3]

Question 17

In ΔABC, points P, Q, and R are chosen so that $AP = AQ$ and $CP = CR$.

Determine the value of x.



$\hat{A} = 180° - 2y$ ISOS Δ: AB = AQ

$\hat{C} = 180° - 2w$ ISOS Δ: PC = RC

$180 - 2y + 180 - 2w = 90°$ (∠s of Δ ABC)

$270° = 2y + 2w$

$135° = y + w$

$y + x + w = 180°$ (str line at P)

$135° + x = 180°$

$x = 45°$

[5]

Alternative solution:

$\hat{Q}_1 = 180° - y$ str line

$\hat{R}_1 = 180° - w$ str line

$x + y + w = 180°$ str line

$\therefore y + w = 180° - x$

In Quad PQBR:

$180° - y + 180 - w + x + 90° = 360°$

$x - y - w = -90°$ (∠s of quad)

$x - (y + w) = -90°$

$x - (180° - x) = -90°$

$2x = 90°$

$x = 45°$